

Rocky Linux 10.1 VM Images – Technical Installation and Lab Guide

1. Overview

Rocky Linux is a community-supported enterprise Linux distribution designed to provide binary compatibility with Red Hat Enterprise Linux (RHEL).

Rocky Linux 10 is the current major release line. Rocky Linux 10 was released on June 11, 2025, and the LinuxVMImages page currently provides ready-to-use Rocky Linux 10.1 virtual machine images for:

- VMware
- Oracle VirtualBox
- Minimal installation
- Graphical installation

The LinuxVMImages page states that Rocky Linux 10 is intended to be 1:1 binary compatible with RHEL 10 and lists support through May 31, 2035.

For production environments, however, administrators should obtain operating-system installation media and packages from official Rocky Linux sources and validate the OS image, packages, and configuration according to their organization's security standards.

2. Rocky Linux 10.1 VM Image Details

The LinuxVMImages Rocky Linux 10.1 page provides the following prebuilt VM configuration.

Component	Configuration
Operating System	Rocky Linux 10.1
VM Platforms	VMware / VirtualBox
Installation Types	Minimal / Graphical
vCPU	2
RAM	2 GB
Virtual Disk	512 GB
<code>/boot</code>	1 GB
<code>/</code>	70 GB LVM
<code>/home</code>	440 GB LVM
Swap	2 GB

Component	Configuration
Network	NAT
Default Username	rockylinux
Default Password	rockylinux

Important

The above configuration is the configuration of the **prebuilt VM image**, not a general Rocky Linux production sizing recommendation.

For example, the 512 GB virtual disk is divided primarily between `/` and `/home`. That layout is convenient for a general-purpose VM but is **not an ideal database-server layout**.

For PostgreSQL or other database workloads, separate mount points for data, WAL, backups, and archive storage are normally more appropriate.

3. Available VM Images

The page provides two primary installation types.

3.1 Minimal Installation

The minimal image is approximately:

- VirtualBox: 828 MB
- VMware: 922 MB

The minimal installation is generally preferable for server and DBA laboratories because it avoids unnecessary graphical packages and reduces the software footprint.

Typical use cases include:

- PostgreSQL server
 - SQL Server on Linux
 - Web servers
 - Application servers
 - Kubernetes/container laboratories
 - Ansible laboratories
 - Linux administration training
 - Database administration training
-

4. Graphical Installation

The graphical image is approximately:

- VirtualBox: 1.8 GB
- VMware: 1.8 GB

The graphical version is useful when the VM is intended for:

- Linux desktop learning
- GUI-based administration
- Desktop application testing
- Linux training for beginners
- Testing graphical administration tools

For a database server lab, the minimal installation is generally simpler because the server does not need a desktop environment.

5. Download Sources

The LinuxVMImages page currently provides download options through SourceForge and another download provider.

  Rocky Linux 10.1 VM Images – LinuxVMImages  <https://www.linuxvmimages.com/images/rockylinux-10/> 

The official Rocky Linux documentation recommends downloading Rocky Linux installation media from the Rocky Linux project and provides the official ISO naming convention.

  Official Rocky Linux Documentation  <https://docs.rockylinux.org/> 

For production deployments, prefer the official Rocky Linux distribution media rather than relying on a third-party prebuilt VM image.

6. Image Integrity Verification

The LinuxVMImages page publishes MD5 and SHA-1 checksums for its VM images. For example, it lists separate checksums for:

- VirtualBox Minimal
- VMware Minimal
- VirtualBox Graphical
- VMware Graphical

Checksum verification is important because it helps determine whether the downloaded file matches the file published by the provider.

However, a checksum alone does not establish that the image is trustworthy or suitable for production.

Recommended approach

For production systems:

1. Download the official Rocky Linux ISO.
 2. Obtain the official checksum/signature information.
 3. Verify the downloaded ISO.
 4. Perform a clean installation.
 5. Apply organizational security hardening.
 6. Register/configure required repositories.
 7. Apply current updates.
 8. Configure monitoring and backup.
 9. Perform vulnerability scanning.
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7. Rocky Linux 10 Major Changes

Rocky Linux 10 introduces several important changes compared with earlier Rocky Linux releases.

According to the official Rocky Linux 10 release documentation, x86-64 systems now require the x86-64-v3 microarchitecture baseline. Earlier x86-64-v2 and older systems are not supported.

Rocky Linux 10 also supports several architectures, including:

- x86-64-v3
- AArch64
- RISC-V
- ppc64le
- s390x

Rocky Linux 10 also removes 32-bit compatibility packages for x86-64 systems.

VMware/VirtualBox implication

Before creating a Rocky Linux 10 VM, verify that the physical host CPU and virtualization configuration satisfy the x86-64-v3 requirement.

This is especially important when using:

- Older physical computers
 - Older VMware hosts
 - Older VirtualBox hosts
 - Nested virtualization
 - Older cloud instances
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8. Rocky Linux 10 Kernel

Rocky Linux 10 uses the Linux 6.12 kernel series as its default kernel baseline.

This is important for administrators because the kernel affects:

- Hardware compatibility
- Networking
- Storage
- CPU scheduling
- Virtualization
- Security
- Container workloads
- Performance

After installing the VM, verify the running kernel with:

```
uname -r
```

For additional operating-system information:

```
cat /etc/os-release
```

Example:

```
NAME="Rocky Linux"  
VERSION="10.1"
```

The exact output can vary as updates are applied.

9. Initial Login

The LinuxVMImages page provides the following credentials for its prebuilt image:

```
Username: rockylinux  
Password: rockylinux
```

The page also indicates that root access can be obtained with:

```
sudo su -
```

Security Recommendation

These credentials should be considered **temporary lab credentials only**.

Immediately after importing the VM:

```
passwd
```

Change the password.

For a production-like environment, create a dedicated administrative account:

```
sudo useradd -m dbaadmin  
sudo passwd dbaadmin
```

Grant administrative privileges according to your organization's policy.

10. First Tasks After VM Import

After importing the VM, perform the following checks.

Step 1 – Check OS

```
cat /etc/os-release
```

Step 2 – Check kernel

```
uname -r
```

Step 3 – Check hostname

```
hostnamectl
```

Step 4 – Check CPU

```
lscpu
```

Step 5 – Check memory

```
free -h
```

Step 6 – Check disks

```
lsblk
```

Step 7 – Check filesystem usage

```
df -h
```

Step 8 – Check IP address

```
ip addr
```

Step 9 – Check NetworkManager

```
systemctl status NetworkManager
```

Step 10 – Check SELinux

```
getenforce
```

These commands provide a quick baseline of the VM before additional software is installed.

11. Update the Operating System

After importing the VM, update the operating system.

```
sudo dnf update -y
```

Then reboot if required:

```
sudo reboot
```

After reboot:

```
uname -r
```

and:

```
dnf updateinfo summary
```

The exact update results depend on the repositories and packages available at the time the command is executed.

12. Disk Layout Analysis

The LinuxVMImages VM uses approximately:

```
512 GB
|
+-- /boot      1 GB
|
+-- /          70 GB LVM
|
+-- /home      440 GB LVM
|
+-- swap       2 GB
```

This is a reasonable generic desktop/server lab layout.

However, it should **not automatically be copied into a PostgreSQL production architecture**.

13. Recommended PostgreSQL Lab Layout

For PostgreSQL administration training, a more useful layout would be:

```
/
├─ /boot
├─ /var
├─ /pgdata
├─ /pgwal
├─ /pgarchive
├─ /pgbackup
└─ /tmp
```

For example:

Mount Point	Purpose
<code>/boot</code>	Boot files
<code>/</code>	Operating system
<code>/pgdata</code>	PostgreSQL data directory
<code>/pgwal</code>	PostgreSQL WAL
<code>/pgarchive</code>	Archived WAL
<code>/pgbackup</code>	PostgreSQL backups
<code>/tmp</code>	Temporary operating-system files

The actual sizing should be based on:

- Database size
- Transaction volume
- WAL generation rate
- Backup retention
- Archive retention
- Recovery objectives
- Growth rate

14. LVM Considerations

The supplied VM uses LVM for `/` and `/home`.

Check the LVM configuration with:

```
lsblk
```

and:

```
sudo pvs
```

```
sudo vgs
```

```
sudo lvs
```

Typical output will show:

Physical Volume
Volume Group
Logical Volume

LVM is useful because logical volumes can be extended without redesigning the entire disk layout.

For database servers, LVM can be used to provide separate logical volumes for database-related storage, subject to the organization's storage and performance design.

15. Network Configuration

The supplied LinuxVMImages configuration uses:

Network: NAT

NAT is convenient for a personal lab because the VM can access external networks through the host.

However, NAT may not be ideal for multi-server database laboratories.

For example, a PostgreSQL HA lab may require:

```
PostgreSQL Server 1
    |
    | Private Network
    |
PostgreSQL Server 2
    |
    |
Backup Server
```

In VMware or VirtualBox, you may therefore use:

- NAT for Internet access
- Host-only/private network for lab communication

or:

- Bridged networking when the VM needs to appear directly on the physical LAN

16. Example PostgreSQL Lab Network

A simple lab could use:

Server	IP Address	Purpose
pgdb01	192.168.10.101	PostgreSQL Primary
pgdb02	192.168.10.102	PostgreSQL Standby
pgbackup01	192.168.10.103	Backup Server

This is only an example private lab design.

Do not use these addresses blindly in an existing network.

17. Hostname Configuration

Set a meaningful hostname:

```
sudo hostnamectl set-hostname pgdb01
```

Verify:

```
hostnamectl
```

Then check:

```
hostname
```

For a multi-server PostgreSQL environment, consistent hostnames are important for:

- Streaming replication
 - Monitoring
 - Backup systems
 - Ansible
 - DNS
 - SSH
 - Troubleshooting
-

18. Time Synchronization

Database servers should maintain accurate system time.

Check:

```
timedatectl
```

Check chrony:

```
systemctl status chronyd
```

Enable it if necessary:

```
sudo systemctl enable --now chronyd
```

Verify synchronization:

```
chronyc tracking
```

Accurate time is important for:

- Database logs
- Backup records
- Replication troubleshooting
- Monitoring
- Incident investigation
- Cluster operations

19. SELinux

Rocky Linux uses SELinux as an important security mechanism.

Check its status:

```
getenforce
```

Possible values include:

```
Enforcing  
Permissive  
Disabled
```

For production systems, do not simply disable SELinux because an application reports a permission problem.

Instead:

1. Identify the denial.
 2. Check SELinux audit logs.
 3. Determine the correct security context.
 4. Configure the required policy/context.
 5. Retest the application.
-

20. Firewall

Check the firewall:

```
sudo systemctl status firewalld
```

List active zones:

```
sudo firewall-cmd --get-active-zones
```

For PostgreSQL, TCP port 5432 may need to be opened when remote clients or replication connections are required.

Example:

```
sudo firewall-cmd --permanent --add-port=5432/tcp
sudo firewall-cmd --reload
```

Do not open database ports broadly to the network.

Restrict access to the required source networks wherever possible.

21. Installing PostgreSQL

Once the Rocky Linux VM has been prepared, PostgreSQL can be installed according to the required PostgreSQL version and supported repository.

Before installation, confirm:

```
cat /etc/os-release
```

```
uname -r
```

```
free -h
```

```
lsblk
```

```
df -h
```

This establishes the baseline before the database installation.

For production, use the PostgreSQL-supported installation method for the required version and validate compatibility with Rocky Linux 10.

22. Why This VM Image Is Useful for DBA Training

A Rocky Linux 10 VM provides a useful environment for practicing enterprise Linux administration alongside database administration.

A DBA can use the VM to practice:

Linux Administration

- `dnf`
- `systemd`
- users and groups
- permissions
- SSH
- `firewalld`
- SELinux
- LVM
- XFS
- networking
- DNS
- `chrony`

PostgreSQL Administration

- PostgreSQL installation
- configuration
- database creation
- roles
- authentication
- backups
- WAL
- replication
- monitoring

- performance tuning
- upgrades
- pgBackRest
- troubleshooting

Automation

- Bash
- Python
- Ansible
- Git
- CI/CD

This makes Rocky Linux particularly useful when learning enterprise PostgreSQL administration on a RHEL-compatible platform.

23. Production vs Lab Usage

There is an important distinction between a ready-made VM image and a production server.

Area	Ready-made VM	Production Server
Installation	Preconfigured	Controlled installation
Credentials	Predefined	Unique credentials
Disk layout	Generic	Workload-specific
Network	NAT	Enterprise network
Security	Requires review	Security baseline required
Packages	Preinstalled	Approved package set
Monitoring	Usually absent	Required
Backup	Usually absent	Required
Hardening	Must be performed	Mandatory
Change control	Not applicable	Required
Documentation	Limited	Required
Vulnerability scanning	Must be performed	Required

Therefore, the image should be treated as a **lab accelerator**, not automatically as a production-ready server.

24. Important Security Checklist

After importing the image:

- [] Change default password
- [] Create administrative account
- [] Review sudo permissions
- [] Update packages
- [] Review enabled services
- [] Review listening ports
- [] Configure firewall
- [] Verify SELinux status
- [] Configure time synchronization
- [] Configure hostname
- [] Configure DNS/hosts
- [] Configure SSH securely
- [] Remove unnecessary packages/services
- [] Scan the VM for vulnerabilities
- [] Configure monitoring
- [] Configure backups where required

Do not expose a newly imported VM directly to the Internet before performing these checks.

25. VM Resource Recommendation for PostgreSQL Training

The supplied image has:

2 vCPU
2 GB RAM
512 GB disk

This is sufficient for basic Linux and small PostgreSQL exercises.

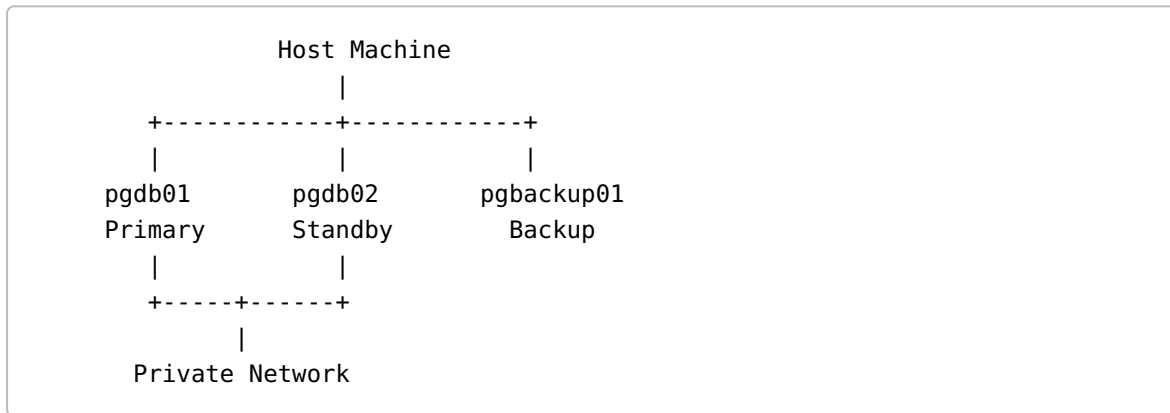
For more advanced PostgreSQL labs, consider:

Lab	vCPU	RAM	Storage
Basic Linux	2	2 GB	40–60 GB
PostgreSQL Basic	2	4 GB	60–100 GB
PostgreSQL DBA Lab	4	8 GB	100–200 GB
PostgreSQL HA Lab	4	8 GB each	100+ GB each
Performance Lab	4–8	16 GB+	SSD/NVMe preferred

These are practical lab starting points, not vendor-required minimums.

26. Recommended Lab Architecture

For serious PostgreSQL DBA practice, a multi-VM design is more useful than a single VM.



You can then practice:

- Streaming replication
- Failover
- Backup and restore
- WAL archiving
- pgBackRest
- Monitoring
- Disaster recovery
- Connection testing
- Network troubleshooting

27. Rocky Linux 10 and PostgreSQL DBA Learning

For someone preparing for enterprise PostgreSQL DBA work, Rocky Linux 10 is particularly useful because it provides exposure to the RHEL-style administration model.

Important commands to become comfortable with include:

```
dnf
systemctl
journalctl
firewall-cmd
getenforce
semanage
restorecon
lsblk
pvs
vgs
lvs
mount
```

```
df
du
ss
ip
nmcli
hostnamectl
timedatectl
chronyc
```

These commands form an important part of day-to-day Linux database administration.

28. Rocky Linux 10 Installation Guidance

Rocky Linux's official installation documentation provides a full server installation procedure and recommends obtaining the ISO from the Rocky Linux project.

For a fresh installation:

1. Download Rocky Linux ISO
2. Verify the ISO
3. Create VM
4. Configure CPU/RAM
5. Configure storage
6. Configure network
7. Boot ISO
8. Start installer
9. Configure storage
10. Configure hostname/network
11. Create administrator
12. Configure software selection
13. Install OS
14. Reboot
15. Update packages
16. Harden OS
17. Configure monitoring
18. Install required applications

Rocky Linux 10 does not support an in-place major-version upgrade from Rocky Linux 8/9 to 10; the official documentation recommends a fresh installation for that transition.

29. Key Takeaways

The LinuxVMImages Rocky Linux 10.1 page provides ready-to-use VMware and VirtualBox images that can significantly reduce the time required to build a Rocky Linux laboratory.

The supplied image provides:

- Rocky Linux 10.1
- VMware support
- VirtualBox support
- Minimal installation
- Graphical installation
- 2 vCPU
- 2 GB RAM
- 512 GB disk
- LVM-based storage
- NAT networking
- Preconfigured lab credentials

However, the image should be treated as a **convenient laboratory starting point**.

For production database environments, build the operating system from approved sources, verify the installation media, apply organizational security standards, design storage according to the database workload, configure networking appropriately, and perform full validation before deployment.

30. Recommended References

LinuxVMImages – Rocky Linux 10.1 VM Images

 [Rocky Linux 10.1 VM Images](https://www.linuxvmimages.com/images/rockylinux-10/) 

Rocky Linux Official Documentation

 [Rocky Linux Documentation](https://docs.rockylinux.org/) 

Rocky Linux 10 Release Notes

 [Rocky Linux 10 Release Documentation](https://docs.rockylinux.org/latest/releases/release_notes/10_0/)  

Rocky Linux 10 Installation Guide

 [Installing Rocky Linux 10](https://docs.rockylinux.org/guides/installation/) 

Final Recommendation

For a **PostgreSQL DBA practice environment**, I would use the LinuxVMImages Rocky Linux 10.1 **Minimal VMware image** as a convenient starting point, then rebuild the storage/network/security configuration to match a real enterprise PostgreSQL server.

For **production**, use the official Rocky Linux installation media rather than treating this third-party prebuilt VM as production-ready.